Mohammad Yazdani

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EDUCATION

UNIVERSITY OF TEHRAN — Tehran, Iran
M.Sc., Process Engineering — GPA: 3.65/4 — *Sep 2023 – Oct 2026*

Thesis: *Surrogate-Assisted Multi-Objective Optimization of an Integrated Adsorption-Cryogenic CO_2 Capture Process for Steam Methane Reforming Flue Gas*
Software: *swingbed*, a Python library for simulating pressure, vacuum and temperature swing adsorption cycles

IRAN UNIVERSITY OF SCIENCE AND TECHNOLOGY — Tehran, Iran
B.Sc., Chemical Engineering — GPA: 3.63/4 — *Sep 2019 – Sep 2023*

Thesis: *From Model to Market: A Feasibility Study of Acrylic Ester Production*

RESEARCH INTERESTS

• Process Systems Engineering (PSE) • Process Modeling, Simulation & Digital Twins • Mathematical Optimization • Carbon Capture, Utilization & Storage (CCUS) • Techno-Economic & Sustainability Analysis • Artificial Intelligence for Chemical Engineering

PUBLICATIONS

JOURNAL ARTICLES

- **Yazdani, M.**, Fatemi, S., and Fakhroleslam, M. "Surrogate-Assisted Multi-Objective Optimization of an Integrated Adsorption-Cryogenic CO_2 Capture Process for Steam Methane Reforming Flue Gas". **Revised (R1)**, *Journal of CO_2 Utilization*. **(Quartile: Q1; IF: 8.6)**
- **Yazdani, M.**, Fatemi, S., and Fakhroleslam, M. "Techno-Economic Assessment of an Optimized Integrated Adsorption-Cryogenic CO_2 Capture Process for Steam-Methane Reforming Flue Gas". **Revised (R1)**, *International Journal of Greenhouse Gas Control*. **(Quartile: Q1; IF: 5.9)**

CONFERENCE ARTICLES

- **Yazdani, M.**, Charoughchi, A., Fatemi, S., and Fakhroleslam, M. "Surrogate-Assisted Multi-Objective Optimization of VPSA for CO_2 Capture from Steam Methane Reforming Flue Gas." The 13th International Chemical Engineering Congress & Exhibition (IChEC 2026), Tehran, Iran, 2026.
- Charoughchi, A., **Yazdani, M.**, Fatemi, S., and Fakhroleslam, M. "Integrated Waste Heat Recovery and Adsorption System for SO_2 Removal from Flue Gas." The 13th International Chemical Engineering Congress & Exhibition (IChEC 2026), Tehran, Iran, 2026.
- **Yazdani, M.**, Charoughchi, A., Fatemi, S., and Fakhroleslam, M. "Simulation of a Hybrid Vacuum Swing Adsorption–Cryogenic Process for CO_2 Removal from Flue Gas." Iranian Conference on Compressed Gases (IGCC), Tehran, Iran, 2025.
- Charoughchi, A., and **Yazdani, M.** "Simulation of Sulfur Dioxide (SO_2) Pollutant Adsorption from High-Sulfur Fuel Combustion." Iranian Conference on Compressed Gases (IGCC), Tehran, Iran, 2025.
- **Yazdani, M.**, Charoughchi, A., Fatemi, S., and Fakhroleslam, M. "Development of Machine Learning-Based Surrogate Models for the Optimization of Direct Air Capture (DAC) Process." 16th International Conference on Soft Computing (ICSC), Tehran, Iran, 2025.
- Charoughchi, A., **Yazdani, M.**, Fatemi, S., and Fakhroleslam, M. "Comparative Study of Response Surface Methodology (RSM) and Random Forest (RF) Models for Predicting SO_2 Recovery from Flue Gas in an Adsorption Process." 6th International Conference on Soft Computing (ICSC), Tehran, Iran, 2025.

RESEARCH EXPERIENCE

UNIVERSITY OF TEHRAN — Tehran, Iran
Research Assistant — *Jun 2024 – Oct 2026*

- Developed Python workflows that automate Aspen Adsorption and Aspen Custom Modeler through the COM interface for process modeling, simulation, optimization, and digital process engineering (GitHub, Medium article).
- Implemented mathematical optimization and AI-enabled surrogate modeling for process design and techno-economic assessment.
- Reduced computational time for process optimization by replacing high-fidelity simulations with surrogate models, enabling rapid exploration of hundreds of operating conditions.
- Developed *swingbed*, a modular Python library for finite-volume simulation of pressure, vacuum and temperature swing adsorption cycles: multicomponent nonisothermal column model with stiff time integration, cycles composed from step and boundary-condition objects, plant scale-up and capture-cost modules, validated against published results and covered by an automated pytest suite.

IRAN UNIVERSITY OF SCIENCE AND TECHNOLOGY — Tehran, Iran
Research Assistant — *Mar 2023 – Sep 2023*

- Led an independent techno-economic and market analysis of five competing industrial production technologies for acrylate esters (Dow, BASF, Toyo Soda, Distillers, Arkema), synthesizing process data, economic modeling, and patent life-cycle analysis into a unified evaluation framework.
- Designed and applied a multi-criteria decision-making methodology (AHP) to systematically compare licensing technologies across economic, market, and technical criteria, resulting in a rigorously justified technology selection.
- Developed a detailed process simulation in Aspen Plus and completed equipment design for the selected process, demonstrating the ability to translate technical literature into a validated, working process model.

TEACHING EXPERIENCE

UNIVERSITY OF TEHRAN — Tehran, Iran
Teaching Assistant – Advanced Engineering Mathematics — *Oct 2024 – Feb 2025*

- Assisted in delivering tutorials and mentoring students in advanced engineering mathematics, guiding analytical and numerical solution methods.
- Developed and delivered hands-on MATLAB workshops covering numerical methods, mathematical modeling, and engineering problem solving.

UNIVERSITY OF TEHRAN — Tehran, Iran
Teaching Assistant – Advanced Adsorption Processes — *Oct 2025 – Feb 2026*

- Assisted in teaching advanced adsorption processes by mentoring students in adsorption theory, process modeling, and simulation concepts.
- Developed and delivered hands-on Aspen Adsorption workshops covering process model development, cycle simulation, and process performance evaluation.

PROFESSIONAL EXPERIENCE

PETRO RAMAN TECHNOLOGY (PRT) CO. — Tehran, Iran
Process Engineer — *Jun 2023 – Present*

- Led the Process Engineering discipline for feasibility, conceptual design, basic and detailed engineering, and debottlenecking studies across petrochemical projects.
- Performed process simulation, engineering calculations, equipment specification, and engineering document review.
- Collaborated with multidisciplinary engineering teams and licensors throughout project execution.

SHAZAND PETROCHEMICAL COMPANY (AR.P.C) Shazand, Iran
Internship *Jul 2022 – Sep 2022*

- Gained hands-on process engineering experience in an operating Olefin unit, working from Technip licensor design documents to perform equipment sizing, hydraulic calculations, and process simulation (Aspen HYSYS/Plus), with direct exposure to live plant operations.

TECHNICAL SKILLS

Programming: Python, MATLAB, FORTRAN

Software Development: Git/GitHub, UI/UX Design

Process Simulation & Engineering Software: Aspen Plus, Aspen HYSYS, AVEVA Process Simulation, DWSIM, UniSim, Aspen Custom Modeler, Aspen Adsorption, Aspen Process Economic Analyzer, Aspen Flare System Analyzer, Aspen Exchanger Design & Rating, HTRI, PIPEFLO, AFT Impulse & Fathom, CONVAL, COMSOL Multiphysics

General: AutoCAD, Microsoft Office, Adobe Products

Languages: Persian (native), English (C1 – TOEFL iBT 100: R27, L25, S24, W24)

HONORS & AWARDS

- Top 1% of participants in the Nationwide University Entrance Examination in Chemical Engineering (B.Sc.)
- Ranked 10th nationwide in the M.Sc. Entrance Examination in Chemical Engineering
- Offered direct admission to the M.Sc. program at Iran University of Science and Technology
- Ranked among the top 20% of undergraduate Chemical Engineering students
- Achieved acceptance in the 28th Student Olympiad of Chemical Engineering

LICENSES & CERTIFICATIONS

AI FOR EVERYONE, BY ANDREW NG Nov 2023
- Issued by DeepLearning.AI (Show credential)

AI PYTHON FOR BEGINNERS, BY ANDREW NG Oct 2024
- Issued by DeepLearning.AI

MACHINE LEARNING SPECIALIZATION, BY ANDREW NG Nov 2024 – Feb 2025
- Issued by DeepLearning.AI, Stanford University (Show credential)

REFERENCES

DR. SHOHREH FATEMI M.Sc. Supervisor
- Professor at Department of Polymer and Chemical Engineering, University of Tehran, Tehran, Iran, shfatemi@ut.ac.ir

DR. MOHAMMAD FAKHROLESLAM M.Sc. Co-Supervisor
- Assistant Professor at Tarbiat Modares University, Tehran, Iran, fakhroleslam@modares.ac.ir

DR. NOROLLAH KASIRI B.Sc. Supervisor
- Professor at Iran University of Science and Technology, Tehran, Iran, kasiri@iust.ac.ir